

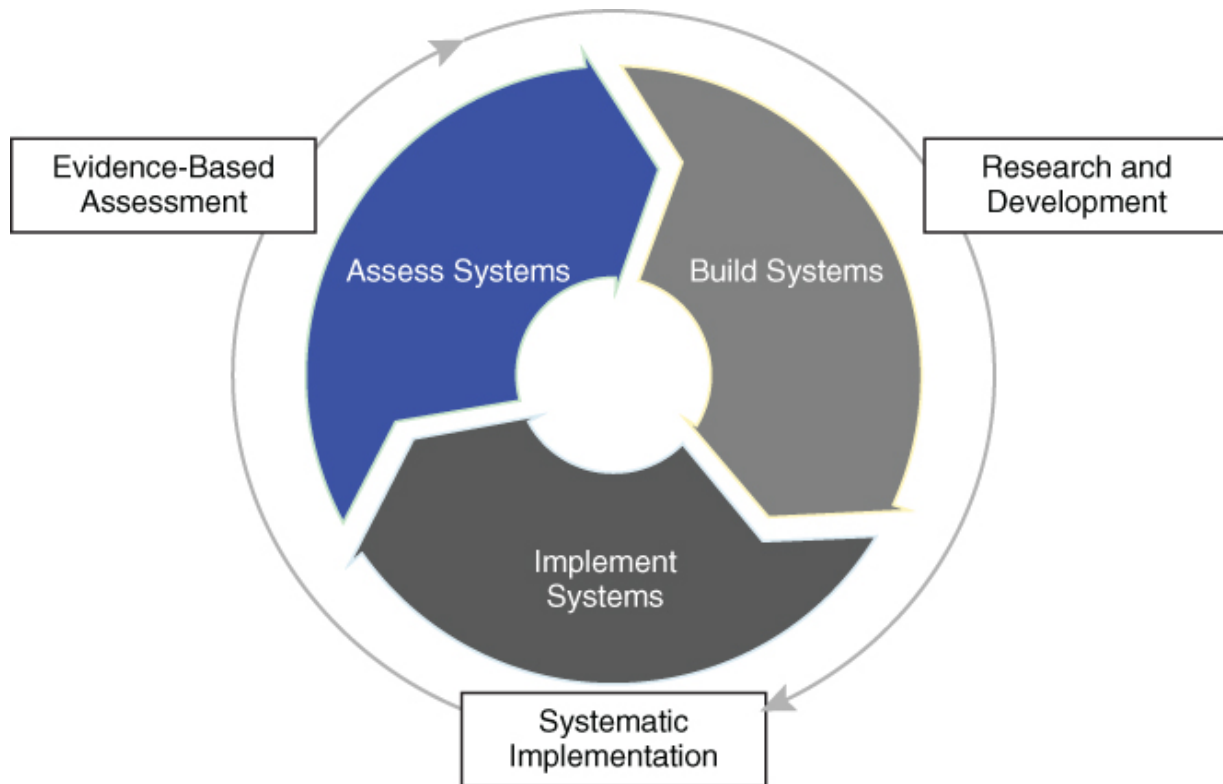


Figure 1.3 The Decision-Making Process

We would argue that human experts are required for the first and third phases of a decision-making process, which are the research and development phase and the assessment phase, respectively. The crux of our argument is that human experts should not be involved in the second phase of decision-making, or the implementation phase.

During the research and development phase of decision-making, experts build and test new ideas. In this phase, experts are required to create a sensible model. In the second phase—implementation—one should eliminate human involvement and rely on systematic execution. Finally, during the assessment phase of decision-making, one should once again rely on human experts to analyze and assess model performance to make improvements and incorporate lessons learned from the implementation phase.

We look to the real world for insights into how this three-phase decision-making framework might be applied in practice. A great case study exists within the US Marine Corps (USMC), where Wes spent nearly four years as an officer deployed in a variety of combat situations. The USMC relies on